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HPRC (High Pressure Ratio Compressor) Testing Results for NEOM Hydrogen Compression

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Abstract

Objective/Scope: NEOM green Hydrogen production project is the first world facility to produce green hydrogen using renewable energy sources. The Hydrogen produced by electrolyzers sees low pressure compression with high flow demand, optimized in power consumption, footprint and Capex/Opex investment. A new technology of moto-centrifugal compressor train called HPRC (High Pressure Ratio Compressor) is the optimum solution to lead all the requirements.

This paper presents the design philosophy, requirements, the predicted performances of HPRC centrifugal compressor and the results obtained in the internal factory test in terms of mechanical and thermodynamic behavior.

Method/Procedure/Process: The presented compression process will be analyzed in the aligned technical solution, HPRC centrifugal compressor for low pressure stages with high flow production and reciprocating compressors for the mid pressure stage. The approach & method to address the specific power availability due to the renewable energy source will be analyzed. More details for each type of compressor are going to be: predicted performance, mechanical behavior in terms of torsional and lateral analysis, and thermodynamic performance. The identified predicted values are compared to internal shop testing results.

Results/Observation/Conclusion: Design predicted performance & internal test results have been defined, shared & agreed with Customer. The intent in the conclusion is:

Emphasize process requirement and challenges due to renewable energy source.

Highlight predicted performance of HPRC solution specifically designed for NEOM project. A new technology of multiple compressor stages in a single casing with a tip speed of 450 m/s.

Compare predicted mechanical & thermodynamic performances to tested values. More detailed analysis on lateral behavior and performance by the closed impeller stages designed specifically for Hydrogen compression.

Besides the above comparison, the conclusion is to demonstrate the benefits proposed in the hypothetical process scheme are confirmed by the predicted performance of centrifugal compression of new technology

Novel/Additive information: The present paper describes the basic process for hydrogen compression defined for NEOM Project. The predicted performance expected for the first innovative HPRC for H2 compression and the results obtained in the internal shop test.

Introduction

Green H2 production, scope and requirements

Green hydrogen, produced through the electrolysis of water using renewable energy sources, is emerging as a cornerstone of the global transition to a low-carbon economy. Its potential to decarbonize hard-to-abate sectors such as heavy industry, transportation, and power generation has positioned it at the forefront of clean energy strategies. However, realizing this potential requires a comprehensive understanding of the production scope, including the scalability of electrolyzer technologies, the availability and integration of renewable energy, and the intended utilization of hydrogen. The electrolyzer technology, renewable profile of the site and the end-use of the hydrogen define a unique production process and set the conditions. Each green hydrogen project must explore, in depth, these technical challenges along with the economic and infrastructural requirements essential for the sustainable and cost-effective deployment of green hydrogen at scale.

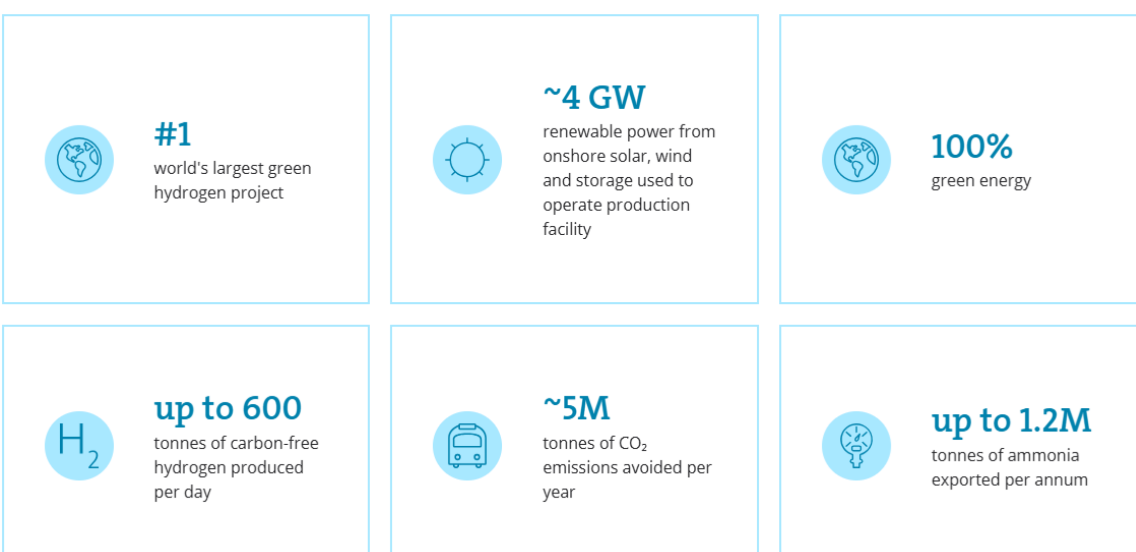
NEOM green H2 production plant

NEOM Green Hydrogen Company (NGHC), an equal production joint venture of ACWA Power, Air Products and NEOM, is establishing the world's largest green-hydrogen-based ammonia production facility run on renewable energy. This mega-plant will produce up to 600 tonnes per day of carbon-free hydrogen in the form of green ammonia as a cost-effective solution for transportation and industrial sectors globally.

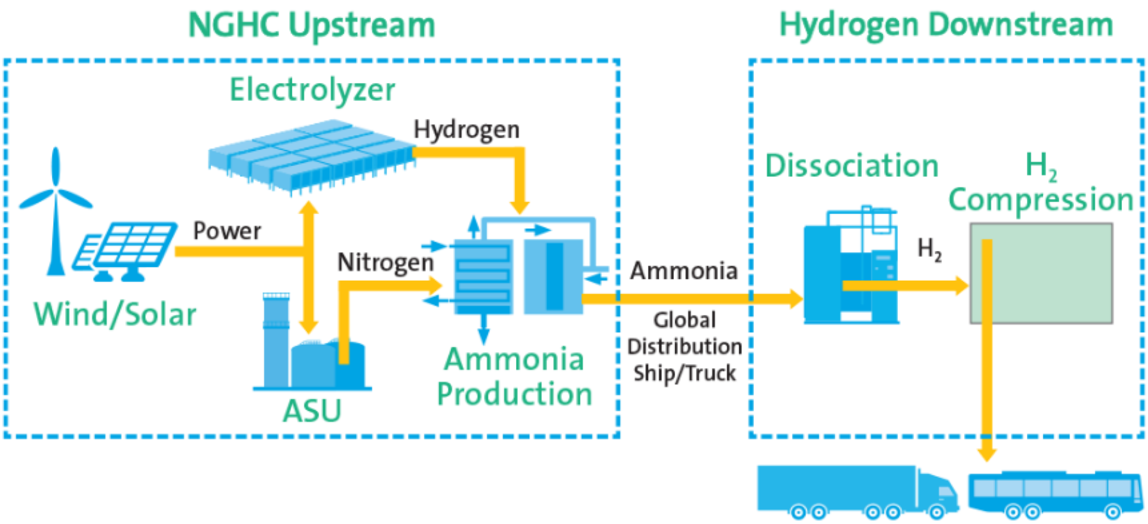
This project is based on proven technologies. Novel aspects include the integration of these technologies, particularly at this scale, and utilizing ammonia to transport hydrogen to global markets. The project will save the world about five million metric tonnes per year of carbon emissions (CO_2).

Plant variable power availability

The 4 GW solar and wind power generation sites are scheduled for completion by mid-2026, with first ammonia product availability expected in 2027



Green hydrogen is produced by electrolysis of water using only renewable energy – solar and wind. Economical green hydrogen production is highly dependent on the cost of electricity. The Kingdom of Saudi Arabia is an ideal location for the NGHC megaproject driven by its abundant sun and wind. Air Products will be the exclusive off-taker of the green products produced at the facility, and NGHC's jetty will be used for the direct transfer of green ammonia to tanker ships, close to global shipping lanes at the crossroads of world distribution.



HPRC compression train solution

To address the unique compression demands of green hydrogen production at NEOM, Air Products has developed a hybrid compression solution that combines the advanced capabilities of Baker Hughes High-Pressure Ratio Compressor (HPRC) with the robustness of reciprocating compressors. The HPRC, a barrel type centrifugal compressor, is the key to this solution with stacked impellers, optimized for efficient operation in low pressure high flow services and operating at high speeds to produce higher pressure ratios. This centrifugal train is optimized for the initial compression of hydrogen directly from electrolyzers, minimizing the number of compression bodies. The compressed gas is fed to the next stage of compression at reduced volume which is then boosted to the required pressure for the ammonia plant by a small number of reciprocating compressors. The multiple HPRC trains respond to the variable hydrogen production with excellent turndown capability and operate to conserve energy. This flexibility to coordinate with the electrolyzers delivers a scalable, energy-efficient, and resilient compression solution tailored to the NEOM project's ambitious sustainability goals. To fully appreciate the elegance of the HPRC compressor, it is essential to understand the fundamentals of this innovative design, walk through the development of the NEOM HPRC solution, and review the factory performance results of the units manufactured for the NEOM project.

HPRC technical solution

Design Principles

The compression ratio that can be achieved by a single compression stage is proportional to the molecular weight MW of the gas to be compressed, to the square power of peripheral tip speed of the impeller U and to the capacity of the impeller to translate kinetic energy into enthalpy increase, namely the work coefficient τ.

By considering a multistage centrifugal compressor, the overall head H that can be delivered is in turn proportional to the number of impellers n, the square power of peripheral tip speed U, the molecular weight MW and the impeller work coefficient τ.

Previous equations suggest that whenever it is required to increase head (or compression ratio) or the same head must be achieved with a lighter gas, such as hydrogen, the number of impellers is likely to increase, potentially resulting in an increase of overall compression bodies with consequent CAPEX and footprint rise. As an alternative, or together with that, the impeller tip speed could be increased, but both design actions will encounter machine configuration and rotordynamic limits.

To minimize the required number of compression stages, hence compression bodies, both stage design and machine configuration must be improved to maximize impeller tip speed and ability to produce compression work.

HPRC technology solves the challenge by means of a combination of key features that allows the rotor to spin faster while increasing impeller peripheral tip speed above those values that represent technological limits for a conventional shrink fit rotor assembly:

- Built-up rotor architecture with impellers stacked by a toothed axial connection and preloaded by a tie-rod which ensures torque transmission throughout the rotor.
- High speed and high efficiency design of compression stages characterized by closed type impellers, low aerodynamic stiffness and optimized mechanical and aeromechanical design.
- Flexure pivot type journal bearings which allow higher peripheral speed and improved rotordynamic behavior with respect to common tilting pad bearings when operating across multiple lateral critical frequencies.

Over the last decade, the HPRC technology has undergone a comprehensive validation and consolidation process, including extensive testing at both component (e.g., impellers and bearings) and system levels, up to full-speed and full-load conditions of complete compression string [1]. Strong confidence has been built in the detailed design and manufacturing quality, particularly in terms of assembly precision and rotor balance thus evolving the technology to be applied to any compressor size and to a broader range of applications, including natural gas, refinery, petrochemical applications and hydrogen of course [2-5].

Green hydrogen applications require compression of hydrogen produced by electrolyzers from a very low pressure with high volumetric flows. HPRC machine configuration fits perfectly in this context: it allows managing the lightest gas in nature to achieve high pressure ratios by maximizing compressor rotating speed, thus minimizing the required number of impellers and compression bodies.

Compressor and rotor construction

In a standard centrifugal compressor, impellers are assembled by shrink fit onto a solid piece shaft. Beyond blades design and material properties, impeller peripheral speed is limited by the maximum radial displacement at hub before interference is lost between shaft and impeller itself. At the same time, a limit exists to allowable increase of interference fit to avoid over-stressing impeller hub and blades to hub connection. Additionally, in a shrink fit rotor the rotordynamic stiffness is primarily driven by the external diameter of the solid shaft (that means the internal diameter of the impeller hub). On the other hand, impeller compression efficiency is directly affected by the so-called aerodynamics stiffness that can be expressed as the ratio between the minimum flow path diameter and the impeller tip diameter. By reducing the aerodynamics stiffness, the impeller efficiency generally increases but this results in a lower flow path diameter, hence reduced impeller hub diameter, since a minimum material thickness must exist in between flow path and impeller hub. A smaller shaft diameter will be then required to install impellers with a consequent decrease of rotordynamic stiffness.

HPRC technology is built around a stacked rotor configuration to overcome both limitations. Impellers are stacked one to each other by a frontal Hirth coupling that is present on both sides of impellers.

Hirth coupling ensures high precision radial centering of parts and torque transmission capability. Stacked arrangement together with optimized impeller design and material selection allows reaching up to 450 m/s of impeller peripheral tip speed in the present solution, with materials suitable for H2 environment, hence removing the risk of interference loss. Moreover, the rotordynamic stiffness of a stacked rotor is driven by the most external part of the rotor bulk, that includes both the minimum flow path diameter of impellers and Hirth connections external diameter [3,6]. Compared to a solid piece shaft with installed impellers of the same size, it results in an increased rotordynamic stiffness or, on the other hand, in the possibility to achieve higher compression efficiency by adopting impellers with lower rotordynamic stiffness without compromising the rotordynamic stiffness that is crucial to maintain stable and smooth rotordynamic behavior.

HPRC compressors selected for NEOM green hydrogen service therefore incorporates one of the latest stages of development in terms of impeller design: closed-type design is stiffness is coupled with a low aerodynamic stiffness and a full 3D blading throughout a large range of volumetric flow, maximizing impeller head [2].

The typical HPRC rotor stacking includes the drive-end shaft-end as well. The central tie-rod is connected into the shaft end and is provided with Hirth connection as well. Then impellers are stacked up one after each other centering onto the Hirth connections. The central tie-rod is then hydraulically preloaded, and the locking nut is locked onto it. Finally, the opposite drive end shaft end is connected by means of a bolted flange to the first impeller of the rotor, Figure 1.

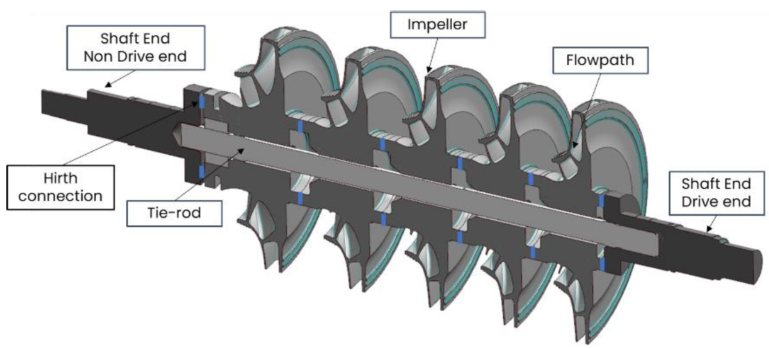


Figure 1—HPRC rotor main items (© 2025 Baker Hughes Company - All rights reserved)

The correct assembly of stacked parts is then checked in details by axial and radial run-out measurements at several rotor sections. Actual rotor profile is recorded and compared with the expected one to ensure required limits are respected. Finally, the rotor is subject to the high-speed balancing operation. The remaining part of the compressor assembly is standard. In case of a barrel machine, as the NEOM compressors, the rotor bundle is assembled and then is fitted into the compressor casing.

Assembled rotors pictures are shown in figure 2.

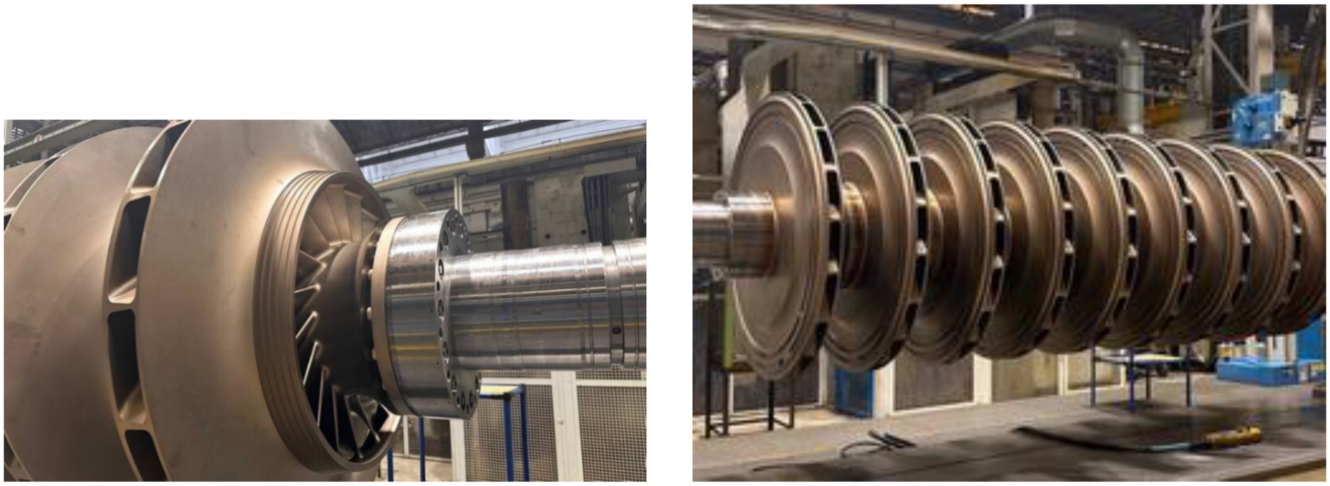


Figure 2—NEOM service compressors assembled rotors (© 2025 Baker Hughes Company - All rights reserved)

Design for variable power availability

The use of renewable energy sources determines an intrinsic discontinuity in terms of power available to the production process. Every time hydrogen production is reduced or stopped the involved turbomachinery, and the centrifugal compressors in particular, are subject to cyclic loads variation which translates into compressors speed ramp up and ramp down. Load and speed variability translate into thermomechanical cycles: the compressor design must therefore include fatigue assessment and develop design strategies to match life requirements in terms of meantime between maintenance (MTBM). Additionally, hydrogen is well known to cause metallic materials properties embrittlement [7-8] determining a significant drop of ductility and toughness that are instead strictly related to the ability of a material to sustain fatigue loads.

Among the different rotary parts of the centrifugal compressors, impeller design requires the primary attention: not only it has to be optimized, as previously described, to maximize impeller peripheral speed while ensuring robust aeromechanical and aerodynamic design but must include specific life cycle assessment, Figure 3.

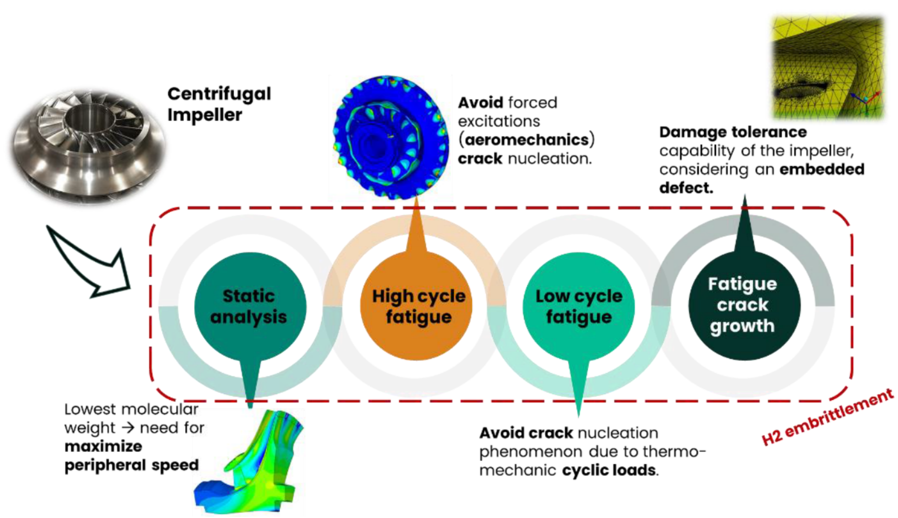


Figure 3—Impeller mechanical design highlights (© 2025 Baker Hughes Company - All rights reserved)

Low cycle fatigue (LCF) analysis is therefore performed to tune impeller design to avoid crack nucleation phenomena due to thermo-mechanical cyclic loads throughout machine lifetime. Different ramp-up and ramp-down of the compressor in terms of time, speed and temperature conditions are checked and the results are used to agree the best option of load cycle range and timing, which ensure the most suitable compromise between rotor loading conditions and plant absorbed power and operability.

Even if LCF analysis can exclude crack nucleation, a fatigue crack growth rate (FCGR) is additionally performed to evaluate the damage tolerance capability of the impellers in presence of a pre-existing or embedded material defect deteriorated by H2 embrittlement material matrix.

Eventually, impeller design results in a multi-purpose technical process where all project's features (equipment minimization targets, flexible plant operability, maintenance intervals timing and adverse gas condition) are effectively addressed at the same time.

HPRC performances prediction

Lateral analysis prediction

Rotordynamic analysis was performed using a state-of-the-art rotordynamic tool (see [9]) and leveraging OEM's internal experience on stacked rotor technology modeling (see [9]). Here below, the example of HPRC1009 analysis is reported. The complete rotor model is shown in Figure 4. The stacked portion of the rotor is modeled as a hollow shaft having a stiffness belt highlighted by green elements. On top of this, the impeller inertias are lumped. The internal tie rod is also modeled to make the analysis very accurate. The rotor supports are modeled through a couple of elements in series: in addition to the usual bearing film (flexure pivot type journal bearings are selected here because of their superior stability performance, see [11]) there are ISFDs. These devices have been extensively studied (see [3]) and they are optimized here to make the first and second mode as much damped as possible.

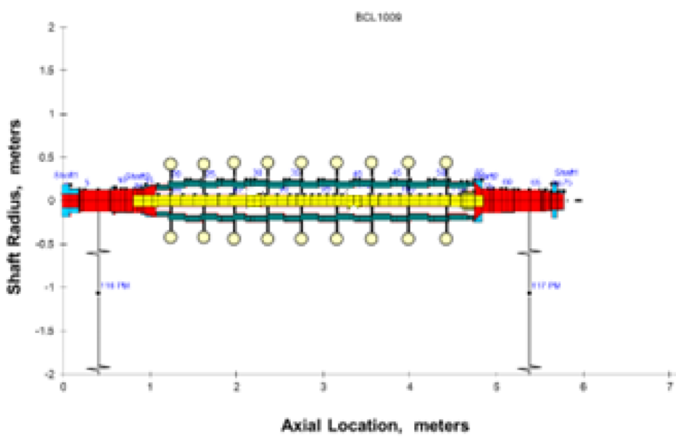


Figure 4—Rotordynamic model (© 2025 Baker Hughes Company - All rights reserved)

Rotordynamic behaviour was finally simulated through the damped unbalance response and the relevant critical speeds detected. Figure 5show both the first critical speed (CS1) and second critical speed (CS2) which are below the operating speed range so they can be detected during MRT. CS1 is expected to be detectable in experimental response, though it is well separate (the relevant AF > 2.5). CS2, on the contrary, is expected to be damped and barely visible in the test (AF < 2.5). CS3 is not shown here but it is expected to be well above MCS so it is not of relevance for the test.

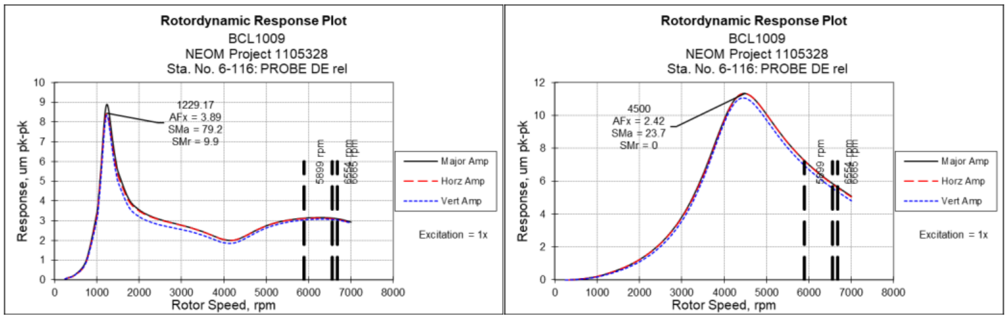


Figure 5—Unbalance response plot relevant to CS1 (left) and CS2 (right) (© 2025 Baker Hughes Company - All rights reserved)

Thermodynamic performances prediction

HPRC rotor design has been designed with specific requirements. In terms of mechanical performance and configuration, the target is to increase the peripherical speed of the stages, without a centered shaft leveraging the frontal and rear teeth connection. In terms of Aerodynamic performances, the intent is to increase stage efficiency and polytropic head, with a design fit for low Mach number.

Mechanical Performance. HPRC compressor is designed with T52 stages, defined to operate at high peripheral speed, above 450 m/s with the use of standard materials. This design target is fulfilled using features like massive hub, tapered blades, and small shrouds. Since no interface with shaft is present and rotor is composed by axial stacking of impellers, the static dimensioning criteria of the wheel is the definition of Burst Speed.

HPRC stages has been designed to guarantee the burst speed definition, here in T52 process design it is worth noting that an optimized behavior of hub blades and shroud that make compliant the radial growth of each substructure, allows the reduction of local strain maximizing the benefit of massive hub and final peripheral speed guaranteed.

Another key step of T52 design is the guarantee of the closure of Hirth serration in the operating speed range. Conversely, from open blades impeller S52 stages to closed blades impeller T52 stages to increase stage efficiency, the radial growth of impeller at Hirth location is higher and the match between front and back Hirth is crucial. Therefore, the hub shoulder at the back side of impeller is dimensioned and placed close to Hirth serration to reduce the radial displacements.

Figure 6 shows a comparison between a standard design stage and a T52 HPRC stage ([12])

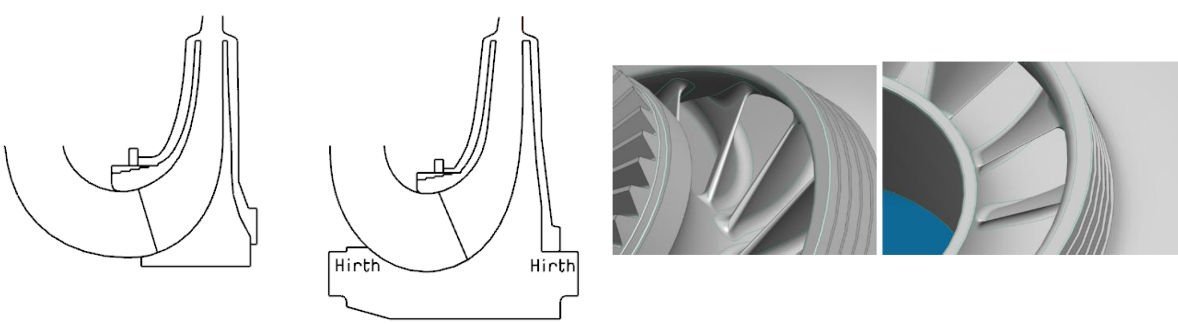


Figure 6—2D view of standard design and T52 HPRC impeller [12] (© 2025 Baker Hughes Company - All rights reserved)

Mechanical design of T52 impeller and associated MCS envelope are based on the analysis of two design conditions:

A. Overload test (IOS) of impeller with material not yet exposed to H2 (no embrittled material)
   Max impeller IOS is based on definition of "burst speed". Burst speed is defined with the minimum speed that verifies the following two criteria:

- Strain limit criterion that prevents impeller from local failure and excessive plastic deformation.
- Stiffness ratio criterion limits the extension of plastic zones in the volume of impeller. It guarantees the containment of centrifugal force and the reversibility of IOS cycle

B. Impeller in steady or transient operating conditions in H2 environment (embrittled material):

Definition of a safe envelope for stress/strain in steady state condition in H_2 environment using dedicated spring load test with increasing load level. Need to consider bi-axial criterion due to relation between stress and H_2 local concentration:

- Understand how material microstructural features (possible H2 traps) interact with hydrogen (Fig.8)
- Multiple test specimens to explore different levels of sVonMises/HPress stress ratio and associate H2 concentration. Rank environmental conditions severity and material performance (Fig.9)
- Define material design space (Fig.10)

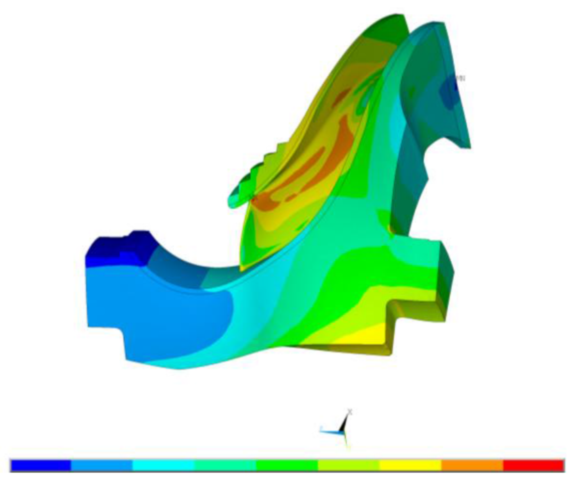


Figure 7—Stress Level at Target MCS Using Standard Steel for Hydrogen Service [12]

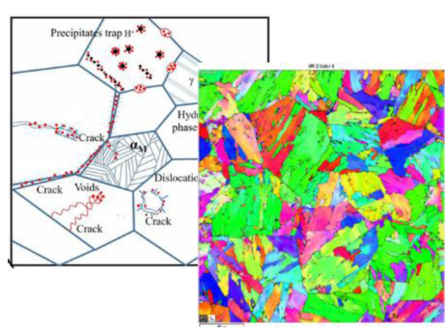


Figure 8—Carbon steel microstructure analysis

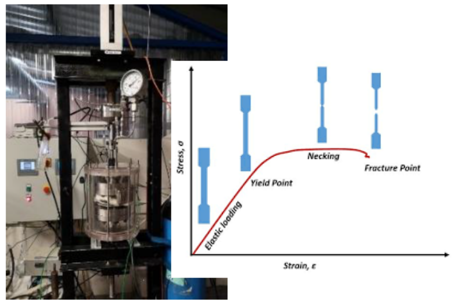


Figure 9—Specimens tests

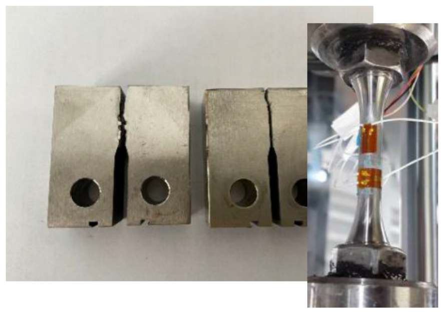


Figure 10—Design space definition tests

Aerodynamic Performance. The stages design has been started in 2019 and completed in 2021. During the development phase the validation has been completed for Performances, Aeromechanical and Manufacturing with dedicated test for each specific area. In term of performances has been tested in different trough model testing where following parameter has been validated:

- Flange to flange performance
- Single components performance (e.g impeller, diffuser, …)
- Flow field and profiles at the intermediate sections
- Stall inception and surge occurrence

The complete T52 has been validated by model test in term of flow coefficient and peripherical Mach number. In the specific case of 100% H2 the Mach number is very low and is close to a specific model test that validate this selection.

In term of performances, T52 stages used for HPRC applications and in this specific case for NEOM centrifugal compressor, have been compared to 3D open impellers and conventional closed impellers. The graph demonstrates an increment of Polytropic efficiency if compared to 2D shrouded impellers (typically used in H2 compression) and 3D unshrouded impellers. T52 design demonstrates also an increase of efficiency compared to 3D shrouded impeller on all the phi stage range.

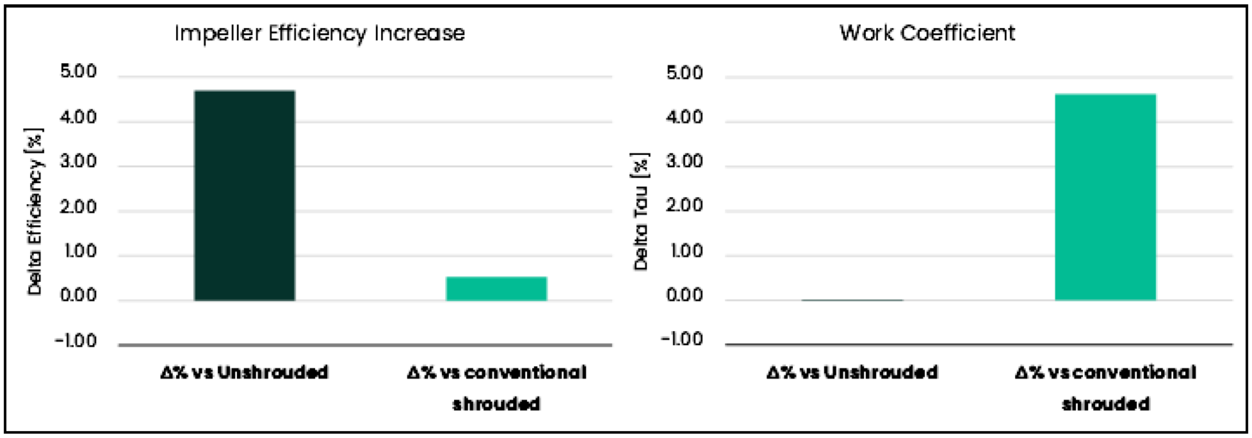


Figure 11—T52 VS unshrouded stages: Polytropic and work efficiency (© 2025 Baker Hughes Company - All rights reserved)

Compressors Expected performances. The expected performance curves are evaluated in the next figure for the single compressor casings. Guarantee point is illustrated by a squared dot on the curve at 100% of speed for the rated case.

Polytropic head demonstrates the wide range from choke line (dotted line in green) to the choke and the choke 10% margin (dotted line in red). Performances are evaluated in six different curves related to six different speeds that cover all the operative range. In term of efficiency curves are overlapped and this demonstrate the efficiency is constant at different speed, but at the same time the efficiency remain constant evaluating at surge and at choke conditions.

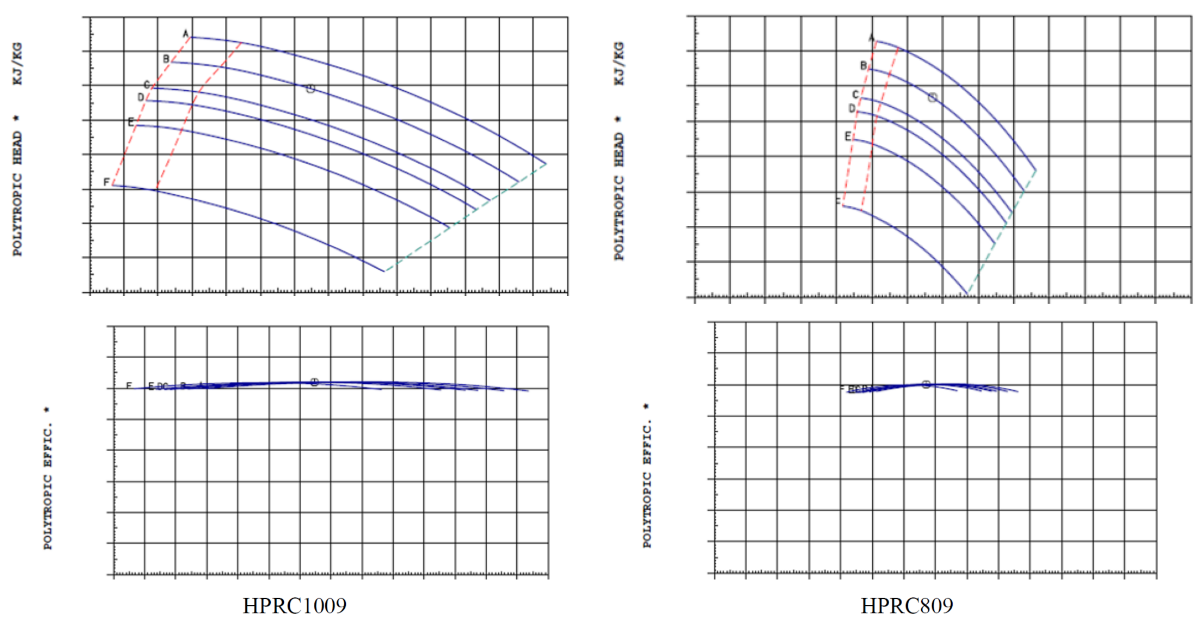


Figure 12—Polytropic head and efficiency expected for HPRC1009 and HPRC809 (© 2025 Baker Hughes Company - All rights reserved)

HPRC testing

Mechanical Running Test Results

The units were finally tested in Florence factory. The first kind of test was Mechanical Running Test (MRT) which allows to validate rotordynamic behavior and general machine mechanical performance. During the test, the shaft vibrations are measured through the proximity probes installed on the bearing housings.

Moreover, the bearing temperatures are monitored through the thermal probes installed into the bearing pads. A typical test sequence includes one run-up run with a steady state condition at MCS and a final run-down. 13 shows the Bode plots relevant to HPRC1009 MRT.

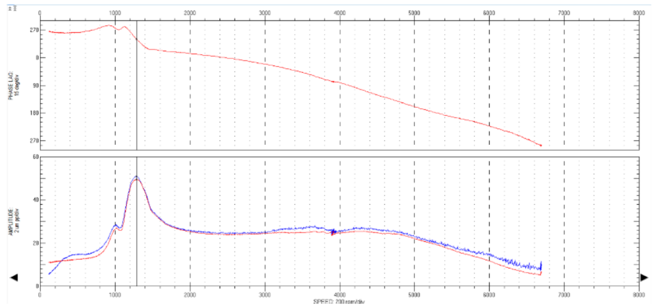


Figure 13—Bode Plots during MRT ()

The rotor transitions through the first critical speed (around 1300rpm) and vibrations are well within the alarm level and then moving to the operating speed range (5900-6800rpm). The second critical speed which is expected to be around 4500rpm is very damped and not clearly detectable. Vibration level is finally well within the acceptance limits (43micron according to customer requirements) and it is almost only due to 1xrev component (e.g. unbalance). This is also confirmed by the waterfall plot showed in which is relevant to the steady state run at MCS. This confirms both rotor and bearings are sound and have stable behavior.

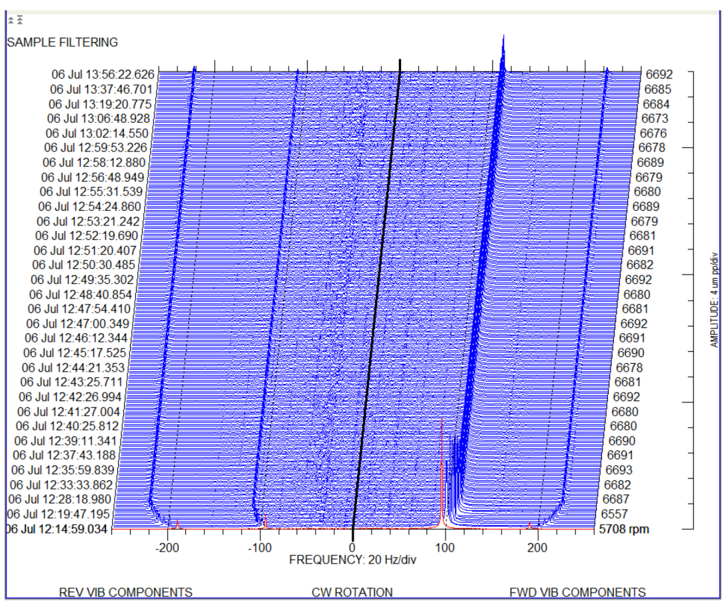


Figure 14—Waterfall plot during steady state run ()

These results from the MRT confirm the sound rotordynamic design and show the relevant predictability is very satisfactory. HPRC809 MRT results are not shown here since they are very similar.

Performance Test of HPRC BCL809 & HPRC1009

The compressor will be factory tested in closed loop, using a shop driver and shop lube oil system. The gas used during the performance test will be composed by 80% Nitrogen (N2) & 20% helium (He) for HPRC1009 compressor and 70% Nitrogen (N2) & 30% helium (He) for HPRC809 compressor.

This gas composition, used for the design of the test, is for reference. During the performance test and according to the actual behavior of the machine, the composition might be adjusted to meet ASME PTC10 type 2 simulated criteria. As per the above, the gas composition may differ among the points recorded for each explored curve.

Gas power is measured by the heat balance method on the process gas (ASME PTC-10, Para. 4.15). Mechanical losses will be measured during the mechanical running test at the maximum continuous speed and converted to rated speed as follows (ASME PTC-10, Para. 5.6.4.):

$$P_{loss}(@rated\,speed) = P_{loss}(@test\,speed)\frac{N^{2.5}{}_{RAI}}{N^{2.5}{}_{TE}}$$

The heat loss from compressor casing will be considered; the heat convection-radiation coefficient is 0,0136 (kW/m2K).

The machine Reynolds number allowable departure is given by ASME PTC 10 - 1997 as shown on ANNEX 11.

During the performance tests, the Reynolds number correction on head and efficiency will be applied according to ASME directives.

The performance test of compressor will consist of five (5) points distributed as follows: one point at overflow, one point between overflow and the guarantee point, one corresponding to the guarantee flow, one point between the guarantee point and surge, one point close to surge. The following are reported test condition for each compressor:

Table 1—HPRC1009 operating conditions ()

Guaranteed operating cond. for similitude design	**U.M.**	**GUARANTEE**	**Type 2 TEST**
Gas molecular weight	-	3.06	23.21
Speed	rpm	6554	2395
Inlet pressure	barA	1.19	1.0
Inlet temperature	°C	42.0	30.0
Discharge pressure	barA	2.90	2.52
Discharge temperature	°C	155.6	153.0
Absorbed power	kW	4327.0	1318

Table 2—HPRC809 operating conditions ()

Guaranteed operating cond. for similitude design	**U.M.**	**GUARANTEE**	**Type 2 TEST**
Gas molecular weight	-	2.48	20.81
Speed	rpm	9022	3179
Inlet pressure	barA	2.73	1.0
Inlet temperature	°C	40.0	30.0
Discharge pressure	barA	6.07	2.32
Discharge temperature	°C	143.0	147.6
Absorbed power	kW	3630.0	501.8

For the surge verification, the flow will be gradually reduced until the minimum flow from the predicted curves is reached and a stable point is recorded. This will validate the contractual operating range. Temperature equilibrium will be established before starting the test readings. Acceptable equilibrium will be demonstrated by six readings for a period of 10 minutes, in which the temperature rise drift does not exceed 5% of temperature rise.

Each test point will consist of three consecutive readings for each instrument after thermal stabilization. Test readings will be averaged to obtain a single value to be used for the computation of results for each measured parameter. Gas composition will be measured and recorded at the start and end of each test.

If a discrepancy in pressure or temperature readings arises during the test, efforts will be made to attempt to correct the abnormal behavior. If the cause of abnormal behavior is a redounded probe, and in case this cannot be corrected during the test, this probe will be removed from the acquisition and the test will continue with the remaining probes. Three correctly functioning probes are the minimum acceptable.

Figure 15—HPRC1009 testing loop (© 2025 Baker Hughes Company - All rights reserved)

The results of the units officially factory tested are illustrated in Figure 19. Black line is reported for reference the 100% speed curve (scaled following ASME PTC10 type 2 similitude criteria for the rated case study. This is the same curve shown in the expected performance curve with the guarantee point. Red curve is the tested curve and the highlighted point is the guarantee contractual point. The left column is referred to polytropic head and efficiency for HPRC1009, the right column is referred to polytropic head and efficiency for HPRC809. All the five point targets to be read in the scope which have been captured following the procedure and performances of the compressor exceed the expected values. This is valid in term pf absolute value but also in terms of operability range. T52 stage with vaneless diffuser demonstrates since the model test, a wide range from choke to surge limit and the test demonstrates that the expected range is confirmed and it is wider when surge has been explored. Choke limit has been determined by evaluating the left limit in terms of analytic values and exploring surge effect (this gives no evidence of synchronous and sub synchronous modes related to surge). The proper design of the stages has been demonstrated by the absolute values of polytropic head and efficiency on middle curve where the percentage of increasing performance is maximized (beyond the guarantee point).

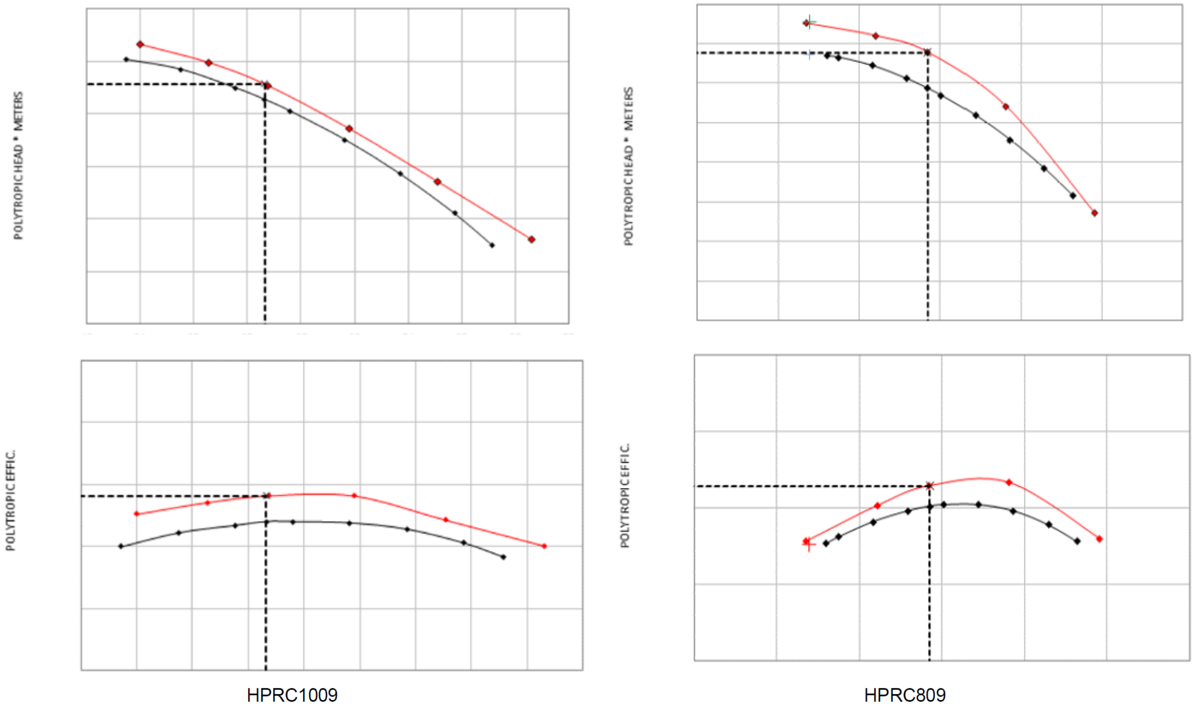


Figure 19—Polytropic head and efficiency as tested (red) vs expected (black) for HPRC1009 and HPRC809 (© 2025 Baker Hughes Company - All rights reserved)

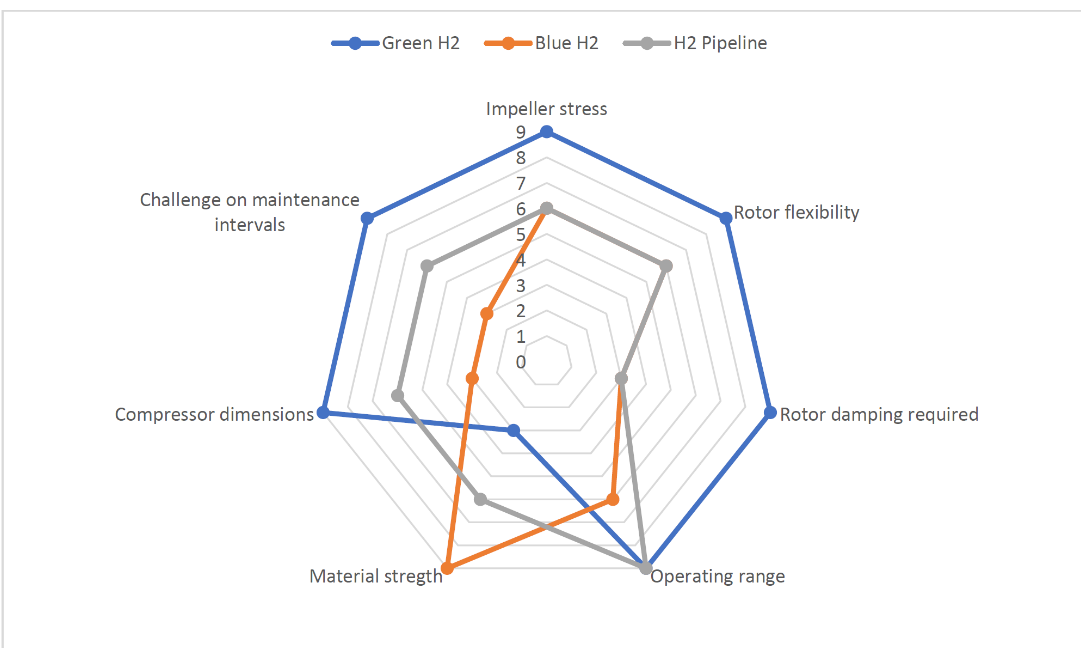


Figure 20—Spider charts of maturity level for Green H2, Blue H2 and H2 pipeline services (© 2025 Baker Hughes Company - All rights reserved)

Results are reported on guarantee point in terms of absolute values for HPRC1009, where the average benefit in terms of efficiency is evaluated to be about 3% increase. Guarantee point conditions have been registered at 125°C and 2.57 barG. At surge, they have been registered at 131°C and 2.68 barG. wherein the inlet pressure and temperature are considered ambient (30°C). In the same way, results are reported on guarantee point in terms of absolute values for HPRC809, where the average benefit in terms of efficiency is evaluated to be about 2% increase. Guarantee point conditions have been registered at 154°C and 1.32

barG. At surge, they have been registered at 164°C and 1.4 barG wherein the inlet pressure and temperature are considered ambient (30°C). All five points analyzed during the test have been stabilized following the defined test procedure and are in line with the expected curve in terms of shape, meaning they are less flat in the left area where the increase of performance is maximized.

Conclusion: a proven platform for the energy transition

The validation campaign presented in this paper has demonstrated the robustness, maturity, and readiness of the HPRC centrifugal compressor technology. Through a multi-year development path, the product has evolved from early-stage prototypes into a fully industrialized solution, capable of meeting the most demanding service requirements in the energy sector.

Each subsystem—rotor dynamics, impeller design, bearing stability, and performance prediction—has been pushed to its operational limits and validated through extensive testing. The rotor has been successfully operated above the fourth critical speed, with modal analysis showing less than 5% deviation between predicted and measured natural frequencies. Impeller tip speeds have exceeded 450 m/s, approached burst conditions, while bearing systems have been validated at surface speeds above 140 m/s, close to their dynamic stability thresholds.

These achievements are not isolated. They are the result of a deliberate and integrated design approach, where predictive tools have been continuously refined to ensure high-fidelity modeling of mass and stiffness matrices, rotor-bearing interactions, and unbalance response across a wide operating envelope. This includes idle conditions, which are critical in hybrid energy systems where renewable availability fluctuates.

The transition from prototype to production has been equally rigorous. A tightly controlled manufacturing process, informed by the validation campaign, has enabled a seamless move into high-intensity production. Performance predictions have been consistently confirmed by test results, underscoring the reliability of the design tools and the repeatability of the manufacturing process.

The spider chart below summarizes the key validation metrics, highlighting the balanced and high-level maturity across all critical design domains:

This holistic and connected design philosophy have elevated the HPRC platform to a new level, one where the compressor is no longer viewed as a collection of components, but as a fully integrated system. Each boundary condition can be stretched and tuned to deliver optimal performance for the specific service environment.

HPRC is now a consolidated product within the Baker Hughes portfolio, with a proven track record and a validated production flow. It is ready for large-scale deployment.

As the energy landscape evolves rapidly in response to climate imperatives, the need for bold, scalable, and reliable solutions becomes urgent. While some sectors may hesitate to embrace change, others are stepping forward as pioneers. When such visionary entities align around a shared mission, transformative outcomes become possible—such as the realization of the world's largest green hydrogen production site, where the HPRC centrifugal compressor stands as a flagship technology.

This is not just a technological milestone. It is a signal that the tools to solve the energy trilemma are within reach—and that with the right mindset, we can deploy them at scale.

References

1. S. Falomi et al (2016). Full Scale Validation Of A High Pressure Ratio Centrifugal Compressor. *Turbomachinery and Pump Symposia*. September 2016, Houston, US.
2. D.T. Rubino, F. Bellobuono, D. Matina, K. Masani, J. Rydland. Application of High-Pressure Ratio Compressor Technology in the Optimized Cascade® Process to Reduce Operational

Greenhouse Gas Emissions and Plot Space. Gastech 2024 Conference. 17-20 September 2024, Houston, US.

3. Vannini, G, Giannelli, C, Matina, D, & Cangioli, F. "High Speed H2 Centrifugal Compressor: a General Review of Journal Bearing Design and Experimental Validation." Proceedings of the ASME Turbo Expo 2024: Turbomachinery Technical Conference and Exposition. Volume 10B: Structures and Dynamics — Fatigue, Fracture, and Life Prediction; Probabilistic Methods; Rotordynamics; Structural Mechanics and Vibration. London, United Kingdom. June 24–28, 2024. V10BT26A005. ASME. https://doi.org/10.1115/GT2024-122482
4. S. Corbò et al (2016). The Role of Turbomachinery in Enabling the Hydrogen Economy. *Adipec*. 11 2021, Abu Dhabi, UAE.
5. D. Berterame, S. Gabriele. (2023). New Compression Solution To Decarbonize Petrochemical Plants. *ASME Turbo Expo*. 06 2023, Boston, US.
6. Venkataraman, B. et al "Dynamics of Modular Rotors in High Speed Centrifugal Compressors: Design, Operational Performance and Field Serviceability." Proceedings of Turbomachinery and Pump Symposia, Texas A&M University, Houston, 2019.
7. Barthélémy, H., 2007, "*Compatibility of Metallic Materials with Hydrogen Review of the Present Knowledge*." [Online]. Available: https://www.h2knowledgecentre.com/content/conference277. [Accessed: 06-Dec-2024].
8. Lee, J. A., and Woods, S., 2016, "*Hydrogen Embrittlement*." [Online]. Available: https://ntrs.nasa.gov/citations/20160005654. [Accessed: 06-Dec-2024].
9. *XLTRC2 Rotordynamic Software Suite*, Texas A&M University Turbomachinery Laboratory: https://turbolab.tamu.edu/wp-content/uploads/2017/10/XLTRC2-NEW.pdf
10. Vannini, G. "Rotordynamic Validation of an Ultra High Speed Multistage Centrifugal Compressor Stacked Rotor." Proceedings of the ASME Turbo Expo 2014: Turbine Technical Conference and Exposition. Volume 7A: Structures and Dynamics. Düsseldorf, Germany. June 16–20, 2014. V07AT31A038. ASME. https://doi.org/10.1115/GT2014-27339
11. Vannini, G, Cangioli, F, Ciulli, E, Nuti, M, Forte, P, Kim, J, & Livermore-Hardy, R. "Experiments on a Large Flexure Pivot Journal Bearing: Summary of Test Results and Comparison With Predictions." Proceedings of the ASME Turbo Expo 2019: Turbomachinery Technical Conference and Exposition. Volume 7B: Structures and Dynamics. Phoenix, Arizona, USA. June 17–21, 2019. V07BT34A009. ASME. https://doi.org/10.1115/GT2019-90278.
12. Lorenzo Toni, Alberto Guglielmo, Angelo Grimaldi, Francesco Cangioli, Davide Biliotti, Elisabetta Belardini, Giulia Meazzini, Lorenzo Miris "Integrated Aero-Mechanical Design and Experimental Validation for Next Generation Compressors for Energy Transition". Proceedings of the ASME Turbo Expo 2023